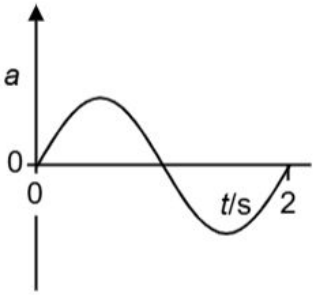
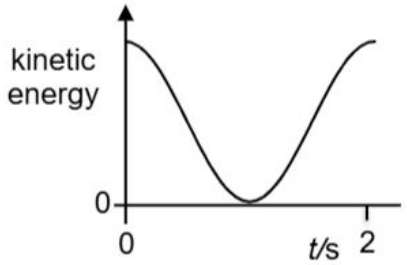
13 A particle oscillates with simple harmonic motion. The graph shows the variation, with time t, of the acceleration a of the particle from $t = 0$ to $t = 2$ s.

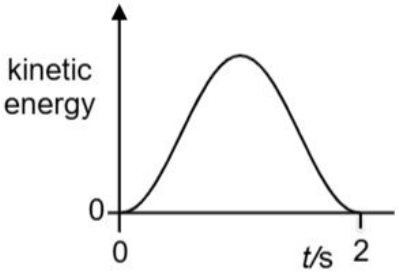


Which graph could represent the variation, with time t, of the kinetic energy of the particle from $t = 0$ to $t = 2$ s?

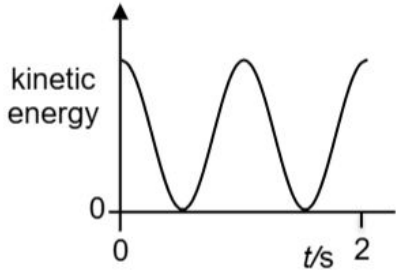
A



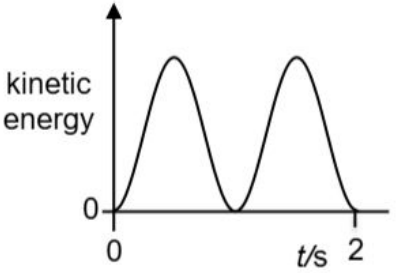
B



C



D



L2 **Answer: C**

At t = 0, acceleration is zero, the oscillating object is at equilibrium position. Kinetic energy is therefore maximum.

In 2 s, acceleration is zero thrice, therefore kinetic energy is maximum thrice.